

SZL Holdings -- Business Proposal

Governed Operational Intelligence for Federal and State Agencies

Prepared for Empire APEX Accelerator

Prepared by: Stephen Lutar, Founder

Date: May 4, 2026

Contact: stephenlutar2@gmail.com

Organization: github.com/szl-holdings

1. Company Overview

SZL Holdings builds governed operational intelligence platforms for regulated enterprises and government agencies. The platform bridges data ingestion, risk analysis, AI-supported decision-making, execution, and auditability into a single continuous workflow.

Every AI-assisted decision the platform makes carries a cryptographic receipt back to the inputs that produced it, the policy that authorized it, and the human who approved it where required. This is not a feature -- it is a property of the runtime.

Legal entity: SZL Holdings, single-member, New York State, in good standing

Founder: Stephen Lutar

Business size: Small business, single-founder

Primary NAICS: 541512 -- Computer Systems Design Services

Secondary NAICS: 541511, 541519, 541611, 541690

2. The Problem

Federal and state agencies are adopting AI systems at accelerating pace. Current procurement frameworks -- NIST AI RMF, DoD Responsible AI Strategy, OMB M-24-10, New York S.B. 7599 -- all require one thing that most AI vendors cannot deliver:

Runtime auditability at the point of decision.

Existing vendors provide AI capabilities (inference, summarization, classification) but ship no mechanism to prove the AI was trustworthy at the moment it made a decision. The audit trail, when it exists, is reconstructed after the fact. The human oversight, when it is claimed, is a checkbox, not a runtime gate.

3. The Solution: A Governed Command Layer

SZL Holdings has built a governed command layer that runs as five continuous stages:

Ingest -- Signals are continuously ingested and normalized across systems (operational, security, financial, document, sensor, third-party).

Score -- Risks and opportunities are identified and scored in real time against policy.

Decide -- Decisions are generated with AI support but require human approval where the policy says they must. This is enforced at the runtime level.

Act -- Actions are executed with full traceability. Every call recorded, every input hashed, every approver named.

Verify -- Outcomes are verified against the original intent and recorded into an auditable history that can be replayed.

The convergence properties of this loop are formalized in the Ouroboros Thesis, a peer-style paper published on Zenodo with DOI 10.5281/zenodo.19944926.

4. Platform Primitives

The platform is built on six primitives that are the same across every vertical:

Outcome Graph -- Every action is a node; every cause is an edge. The graph is the system of record for what was decided and why.

Proof Chain -- Every output carries a hash chain back to the inputs and the human who approved each gated step.

Covenant Policy -- Declarative policy that decides which actions are auto-allowed, which require human approval, and which are forbidden.

Decision Simulation -- Decisions can be replayed against historical state to test policy changes before they ship.

Workflow Engine -- The bounded-loop scheduler that runs the five stages of the governed command layer.

Event Fabric (PRISM Bus) -- The append-only event spine. Everything that happens publishes here.

5. Live Product Surfaces

The platform runs as seven customer-facing product surfaces orchestrated by A11oy:

Product	Domain	Government Relevance
A11oy	Orchestration + Decision Intelligence	AI governance, operational transparency, audit infrastructure
Sentra	Cyber Resilience Command	SOC operations, threat intel, CISA/Zero Trust alignment
Terra	Real Estate Intelligence	State property portfolio management, distress discovery

Vessels	Maritime Intelligence	DHS/Coast Guard maritime domain awareness
Counsel	Legal Matter Command	Agency legal ops, traceable legal operations
Amaru	Convergent Reverse-ETL	Cross-system data reconciliation with auditable lineage
Carlota Jo	Concierge Advisory	Premium service brand

All seven surfaces are live and operational. Screenshots available in the dossier package.

6. Public Proof Artifacts

Two artifacts are public on GitHub today. This is significant for government procurement because it means the claims are independently verifiable:

Ouroboros Runtime (open source)

- Repository: github.com/szl-holdings/ouroboros
- Current release: v6.2.0
- Tests, dependabot, CodeQL, SECURITY.md
- This is the actual code, not a marketing repository

Ouroboros Thesis (DOI-pinned)

- Repository: github.com/szl-holdings/ouroboros-thesis
- Current release: paper-v3-2.0.0
- DOI: 10.5281/zenodo.19944926
- License: CC-BY-4.0
- ORCID: 0009-0001-0110-4173

The May 2, 2026 release publicly retracted and replaced an earlier version after a self-audit identified residual fabrications. This self-correction is in the public record and demonstrates the governance discipline the platform is built to enforce.

7. Government Standards Alignment

NIST AI Risk Management Framework

Function	SZL Coverage
GOVERN	Validator registry, loop policy, operator modes, budget governance
MAP	Domain pack router, task-type routing, risk-tier classification

MEASURE	Delta and consistency scores per step, uncertainty quantification
MANAGE	Human approval gate, halt conditions, replay verification

DoD Responsible AI Tenets

Tenet	SZL Coverage	Status
Responsible	Human approval at risk tier 3/4, validator hard stops	Covered
Equitable	Bias testing plan in development	In Progress
Traceable	Full trace runtime, append-only logs, cryptographic receipts	Covered
Reliable	Golden runs, replay verification, consistency gates	Covered
Governable	Approval gate, halt conditions, operational modes	Covered

New York State AI Procurement Climate

SZL Holdings is positioned for New York State opportunities under:

- NY DIGIT Agency vendor standards for AI adoption across 50+ state agencies
- S.B. 7599 AI Transparency Law -- requiring public disclosure of AI tools and bias mitigation
- OGS centralized contract vehicles for AI governance

8. Target Sectors

Sector	Use Case	Vehicle
DoD Responsible AI	Runtime auditability for AI-in-the-loop decisions	SBIR Phase I, subcontract
DHS / CISA	AI governance under M-24-10 inventory requirements	Sources-sought response
NIST	Reference implementation for runtime trust scoring	Direct engagement
GSA Schedule 70 / MAS IT	Base vehicle for civilian agency deployments	GSA MAS application
NY State OGS	State-level AI governance	Centralized contract

9. Verified Platform Metrics

Every number below is produced by a documented verification command. Re-runnable.

Metric	Value
Customer-facing product surfaces	7 + A11oy orchestration layer
Platform primitives	6
Database tables (provisioned)	848
API endpoint declarations	5,524
Industry verticals	7
Monorepo packages	126
Original innovations	44
Runtime tests passing	133
Security tests passing	126
CI workflows	23

10. What SZL Holdings Is Not Claiming

This section is included deliberately. Honesty builds trust with procurement professionals:

- SZL Holdings is not currently performing on any federal contract
- SZL Holdings has not received a federal cloud authorization, ATO, or DoD impact-level designation
- SZL Holdings has not been audited by an outside firm
- SZL Holdings has no signed customer contracts for the platform at this writing
- SZL Holdings is a single-founder operation
- The numbers above are internal platform metrics -- they describe what has been built, not revenue or users

The strength of the position is the public proof, not pretended traction.

11. 90-Day Roadmap

Timeline	Milestone
Days 1-14	SAM.gov UEI activated, NAICS confirmed
Days 15-30	CAGE code issued, CMMC Level 1 self-assessment posted
Days 31-60	First SBIR opportunity identified with APEX guidance
Days 61-90	First sources-sought response submitted

12. Contact

Stephen Lutar, Founder

SZL Holdings

Email: stephenlutar2@gmail.com

GitHub: github.com/szl-holdings

ORCID: 0009-0001-0110-4173

Attachments

- Capability Statement (one-pager)
- Operational Briefing (detailed)
- Government Readiness Audit
- Product screenshots (9 images)
- Ouroboros Thesis PDF (available at Zenodo DOI)